

BOGUE USMC, NC, USA

WMO: 723097

Lat: 34.683N

Lon: 77.033W

Elev: 21

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	27.1	4.9	7.1	29.0	10.1	9.2	32.7	21.8	63.1	19.1	54.7	5.9	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	89.5	77.7	87.8	77.1	86.3	76.7	80.5	86.0	79.8	85.4	78.8	84.2	10.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	150.1	84.1	78.0	145.6	83.4	77.2	141.7	82.7	44.1	86.4	43.1	85.6	42.3	84.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	17.8	16.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	45.4	49.5	54.4	62.3	70.9	77.2	79.5	77.1	75.1	66.3	54.6	50.3
	DBStd	14.06	11.10	10.71	8.50	6.68	5.69	4.30	4.68	6.23	5.09	7.73	8.36	8.24
	HDD50	556	223	128	53	3	0	0	0	0	0	1	46	102
	HDD65	2416	612	447	340	125	18	0	0	8	4	79	325	458
	CDD50	5526	81	114	188	372	646	816	913	841	754	507	183	111
	CDD65	1913	6	13	10	44	199	367	448	384	307	120	13	2
	CDH74	15990	69	136	66	81	991	3199	4678	3668	2366	702	25	9
	CDH80	4255	20	32	13	6	153	773	1534	1103	537	83	0	1
Wind	WSAvg	7.6	6.7	7.5	8.0	8.8	8.7	8.4	8.4	6.9	7.4	7.0	6.8	6.7
Precipitation	PrecAvg	55.5	4.1	3.7	4.1	3.2	4.1	5.2	6.4	7.1	6.5	3.9	3.4	3.8
	PrecMax	78.9	7.8	10.2	9.9	6.8	9.7	11.2	11.8	14.6	19.5	10.6	9.2	7.2
	PrecMin	37.0	1.8	1.7	1.6	0.6	1.7	2.1	3.3	2.6	1.1	0.1	0.7	1.1
	PrecStd	9.6	1.7	1.9	1.9	1.5	1.9	2.3	2.4	2.8	4.4	2.6	2.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.8	85.1	81.6	78.8	89.0	91.6	92.9	91.4	88.6	85.3	77.1	72.5
		MCWB	65.8	64.4	64.1	70.4	73.4	75.5	78.1	77.4	75.1	74.9	69.9	63.9
	2%	DB	72.9	78.1	73.7	76.3	83.3	88.1	89.3	88.4	86.4	82.2	73.8	68.5
		MCWB	60.5	61.3	61.6	68.0	72.8	75.8	79.1	77.9	75.9	74.5	68.1	64.0
	5%	DB	66.4	73.4	70.3	74.3	80.9	85.9	87.9	86.6	85.0	80.2	71.3	66.5
		MCWB	59.1	62.1	61.6	66.6	72.3	75.5	78.4	77.2	75.7	72.8	66.2	62.9
	10%	DB	62.5	66.1	67.1	72.5	79.0	84.2	86.3	85.2	83.3	78.1	68.5	64.2
		MCWB	57.9	58.6	60.4	65.5	71.0	74.9	77.8	76.6	74.8	71.7	63.5	61.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	69.1	67.5	73.4	77.2	80.3	82.1	81.1	79.7	78.0	72.2	67.2
		MCDB	79.3	79.4	73.4	76.3	83.0	84.8	88.5	85.9	84.7	81.3	75.2	70.3
	2%	WB	63.9	65.6	65.4	70.5	75.2	78.9	81.0	80.0	78.3	76.5	70.0	65.1
		MCDB	69.0	72.0	70.1	74.1	81.1	84.5	86.9	85.5	83.6	80.1	72.8	67.5
	5%	WB	61.5	62.8	63.9	68.6	73.4	78.0	80.2	79.1	77.4	75.2	67.6	63.7
		MCDB	65.7	70.3	68.6	72.6	78.4	83.2	85.9	84.7	83.0	79.1	70.3	65.8
	10%	WB	58.4	59.9	61.9	66.9	72.5	77.2	79.5	78.2	76.3	73.2	64.6	61.5
		MCDB	62.0	65.8	66.3	70.9	77.3	82.0	85.0	83.6	81.6	77.4	67.7	64.0
Mean Daily Temperature Range	5% DB	MDBR	19.2	19.8	17.9	15.7	13.9	12.9	11.5	11.7	13.5	16.2	18.5	17.3
		MCDBR	23.8	28.5	19.1	15.0	13.4	14.1	12.5	12.3	13.3	14.5	18.7	16.9
		MCWBR	16.4	16.0	11.8	8.9	6.7	7.1	6.5	6.2	6.3	8.2	13.2	13.8
	5% WB	MCDBR	20.6	21.9	15.8	11.7	11.7	12.3	11.3	10.5	11.2	11.0	15.6	16.0
		MCWBR	16.6	15.1	11.8	8.2	6.5	7.2	6.6	6.4	6.4	7.5	13.6	14.8
Clear-Sky Solar Irradiance	taub	0.325	0.334	0.362	0.401	0.447	0.491	0.520	0.501	0.423	0.377	0.346	0.333	
	taud	2.534	2.504	2.444	2.331	2.247	2.169	2.102	2.149	2.371	2.476	2.511	2.538	
	Ebn at Noon	280	290	289	281	268	255	247	249	266	271	269	270	
	Edh at Noon	26	30	34	40	44	48	50	47	36	30	26	24	
All-Sky Solar Radiation	RadAvg	830	1062	1405	1773	1955	2023	1910	1737	1440	1224	921	737	
	RadStd	50	101	107	118	132	111	114	103	96	128	65	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+212

Nomenclature: See separate page